

Puzzles to Puzzle You (Remastered)

Welcome to the ninth iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (Knights and Knaves) and pose the next one.

This puzzle was given as the midterm bonus to my summer Symbolic Logic students. All of my students got it correct.

Puzzle #8 Knights and Knaves

On the island of Knights and Knaves, everyone is either a Knave or a Knight. *Knights always tell the truth. Knaves always lie.* Given the following statements, determine for each islander if they are a Knight or a Knave.

1. Piggy says: Ralph always tells the truth.
2. Ralph says: Jack is a Knave.
3. Jack says: Piggy is lying.
4. Jack says: Piggy and I are the same type.

Ralph = Piggy = Knight, Jack = Knave

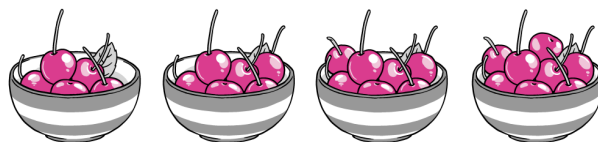
Shoutout (Ivan) I spent a good chunk of my summer learning about Gaussian integers from Ivan, who has a real gift for making obscure, scary math seem intuitive and, in fact, natural. For those of you who have read Elif Batuman's *The Idiot*, the first half of that novel was basically my summer with Ivan.

Solution One way to solve this puzzle is with a truth table. Another way is to reason accordingly: Either Piggy is a knight or a knave. Suppose he is a knave. Then (1) is false so Ralph is a knave. Then (2) is false, so Jack is a Knight. Then (4) is true. Since Jack is a Knight, it follows that Piggy is a Knight. But we assumed initially that Piggy is a Knave! $\perp$ Therefore, Piggy is a knight. By (1), it follows that Ralph is a knight. By (2), it follows that Jack is a Knave. $\square$

Puzzle #9 Splitting Cherries

In front of you and your friend are 4 bowls of cherries, containing, respectively, 5, 6, 7 and 8 cherries. You and your friend will alternately pick a bowl and take one or more cherries from it.

1. If you go first, and you want to be sure to get the last cherry, how many cherries should you take—and from which bowl?
2. If you go first, and you want to be sure your friend gets the last cherry, how many cherries should you take—and from which bowl?



This cherry puzzle is a version of the combinatorial game Nim, in which two players take turns removing (or "nimming") objects from distinct piles. On each turn, a player must remove at least one object, and may remove any number of objects provided they all come from the same pile. Depending on the version being played, the goal of the game is either to avoid taking the last object or to take the last object.

Nim is fundamental to the Sprague–Grundy theorem, which says that every impartial game—a game where the set of allowable moves depends only on the position of the game and not on which of the two players is moving—is equivalent to a nim game with a single pile.